C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 1.01

Functions, Modulus & Quadratics

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

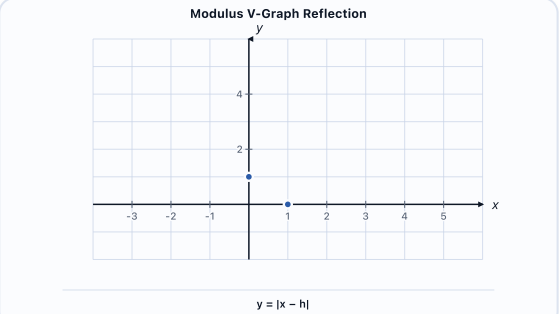
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 C	Q3 A	Q4 A	Q5 D
Q6 A	Q7 B	Q8 C	Q9 A	Q10 B
Q11 C	Q12 D	Q13 B	Q14 D	Q15 A
Q16 B	Q17 B	Q18 C	Q19 D	Q20 A
Q21 C	Q22 A	Q23 B	Q24 A	Q25 D

Functions, Modulus & Quadratics Practice Exercises

1 MCQ

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.



A $f^{-1}(x) = \frac{3x-2}{x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-3}$

C $f^{-1}(x) = \frac{x-2}{3x+1}$

D $f^{-1}(x) = \frac{2x+1}{x-3} \quad (x \neq 3)$ ✓

WORKED SOLUTION & VERIFICATION

Let $y = \frac{3x+1}{x-2} \implies y(x-2) = 3x+1 \implies yx - 2y = 3x+1 \implies x(y-3) = 2y+1 \implies x = \frac{2y+1}{y-3}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-3}$. [M1][A1]

2 MCQ

The function $f(x) = \frac{4x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x-1}{x-4}$

B $f^{-1}(x) = \frac{x-2}{4x+1}$

C $f^{-1}(x) = \frac{2x+1}{x-4} \quad (x \neq 4)$ ✓

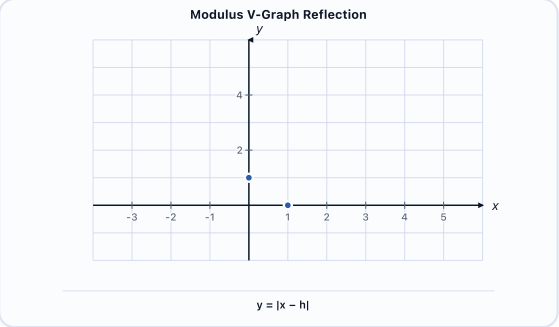
D $f^{-1}(x) = \frac{4x-2}{x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{4x+1}{x-2} \implies y(x-2) = 4x+1 \implies yx - 2y = 4x+1 \implies x(y-4) = 2y+1 \implies x = \frac{2y+1}{y-4}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-4}$. [M1][A1]

3 MCQ

The function $f(x) = \frac{5x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.



- **A** $f^{-1}(x) = \frac{2x+1}{x-5} \quad (x \neq 5)$ ✓
- **B** $f^{-1}(x) = \frac{x-2}{5x+1}$
- **C** $f^{-1}(x) = \frac{5x-2}{x+1}$
- **D** $f^{-1}(x) = \frac{2x-1}{x-5}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{5x+1}{x-2} \implies y(x-2) = 5x+1 \implies yx-2y = 5x+1 \implies x(y-5) = 2y+1 \implies x = \frac{2y+1}{y-5}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-5}$. [M1][A1]

4 MCQ

The function $f(x) = \frac{6x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

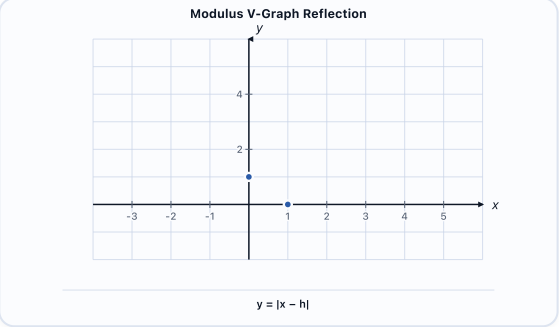
- **A** $f^{-1}(x) = \frac{2x+1}{x-6} \quad (x \neq 6)$ ✓
- **B** $f^{-1}(x) = \frac{x-2}{6x+1}$
- **C** $f^{-1}(x) = \frac{2x-1}{x-6}$
- **D** $f^{-1}(x) = \frac{6x-2}{x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{6x+1}{x-2} \implies y(x-2) = 6x+1 \implies yx-2y = 6x+1 \implies x(y-6) = 2y+1 \implies x = \frac{2y+1}{y-6}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-6}$. [M1][A1]

5 MCQ

The function $f(x) = \frac{7x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.



- **A** $f^{-1}(x) = \frac{7x-2}{x+1}$
- **B** $f^{-1}(x) = \frac{2x-1}{x-7}$
- **C** $f^{-1}(x) = \frac{x-2}{7x+1}$
- **D** $f^{-1}(x) = \frac{2x+1}{x-7} \quad (x \neq 7)$ ✓

WORKED SOLUTION & VERIFICATION

Let $y = \frac{7x+1}{x-2} \implies y(x-2) = 7x+1 \implies yx-2y = 7x+1 \implies x(y-7) = 2y+1 \implies x = \frac{2y+1}{y-7}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-7}$. [M1][A1]

6 MCQ

The function $f(x) = \frac{8x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

- **A** $f^{-1}(x) = \frac{2x+1}{x-8} \quad (x \neq 8)$ ✓
- **B** $f^{-1}(x) = \frac{x-2}{8x+1}$
- **C** $f^{-1}(x) = \frac{2x-1}{x-8}$
- **D** $f^{-1}(x) = \frac{8x-2}{x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{8x+1}{x-2} \implies y(x-2) = 8x+1 \implies yx-2y = 8x+1 \implies x(y-8) = 2y+1 \implies x = \frac{2y+1}{y-8}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-8}$. [M1][A1]

7 MCQ

The function $f(x) = \frac{9x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

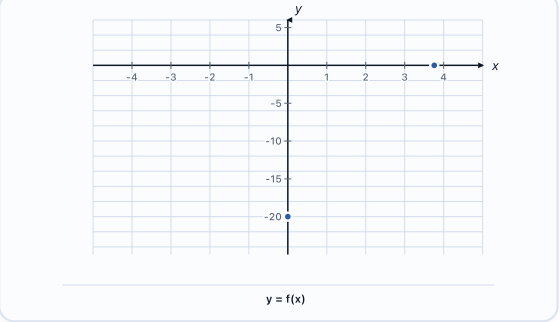
- **A** $f^{-1}(x) = \frac{2x-1}{x-9}$
- **B** $f^{-1}(x) = \frac{2x+1}{x-9} \quad (x \neq 9)$ ✓
- **C** $f^{-1}(x) = \frac{x-2}{9x+1}$
- **D** $f^{-1}(x) = \frac{9x-2}{x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{9x+1}{x-2} \implies y(x-2) = 9x+1 \implies yx-2y = 9x+1 \implies x(y-9) = 2y+1 \implies x = \frac{2y+1}{y-9}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-9}$. [M1][A1]

8 MCQ

Consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. Find the remainder $f(2)$ upon division by $(x-2)$.



- **A** -10
- **B** -7
- **C** -12 ✓
- **D** -22

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** By the Remainder Theorem, $f(2) = 2^3 - 5(2^2) + 10(2) - 20 = -12$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: -12. **[A1]**

9 MCQ

The quadratic graph $f(x) = 4x^2 + gx + 256$ is tangent to the horizontal axis with $g > 0$. Find the positive coefficient g.

- **A** 64 ✓
- **B** 68
- **C** 72
- **D** 54

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Setting discriminant $\Delta = g^2 - 4(4)(256) = 0$ gives $g = 64$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 64. **[A1]**

10 MCQ

Find the set of values of the constant k for which the equation

$$|x^2 - 4| = 2x + k$$

has exactly four distinct real solutions.

A $4 \leq k \leq 5$

B $4 < k < 5$ ✓

C $-4 < k < 4$

D $k > 5$

WORKED SOLUTION & VERIFICATION

To find the number of solutions to $|x^2 - 4| = 2x + k$, we consider the intersections of the curve $y = |x^2 - 4|$ and the line $y = 2x + k$.

The curve is defined piecewise:

- Inner arch: $y = 4 - x^2$ for $-2 \leq x \leq 2$
- Outer branches: $y = x^2 - 4$ for $x < -2$ and $x > 2$

1. **Intersection with the inner arch ($-2 \leq x \leq 2$):**

$$4 - x^2 = 2x + k \iff x^2 + 2x + (k - 4) = 0$$

For two distinct real intersections, the discriminant must be strictly positive:

$$\Delta = 2^2 - 4(1)(k - 4) = 20 - 4k > 0 \implies k < 5$$

At $k = 5$, $\Delta = 0$, so the line is tangent to the inner arch at $x = -1 \in (-2, 2)$, giving only 1 intersection point. Furthermore, the roots must lie within the interval $[-2, 2]$. The lower critical value occurs when the line passes through the corner intercept $(-2, 0)$:

$$0 = 2(-2) + k \implies k = 4$$

At $k = 4$, the equation becomes $x^2 + 2x = 0 \implies x = 0$ or $x = -2$. Because $x = -2$ is the boundary corner shared with the outer branch, the total number of distinct solutions is 3.

For $4 < k < 5$, the roots are $x = -1 \pm \sqrt{5 - k}$. Since $0 < 5 - k < 1$, we have $\sqrt{5 - k} \in (0, 1)$, which guarantees both roots strictly lie in $(-2, 0) \subset (-2, 2)$. Hence, there are exactly 2 distinct solutions on the inner arch.

2. **Intersection with the outer branches ($x < -2$ or $x > 2$):**

$$x^2 - 4 = 2x + k \iff x^2 - 2x - (k + 4) = 0$$

The discriminant is:

$$\Delta = (-2)^2 - 4(1)(-(k + 4)) = 4k + 20$$

For $4 < k < 5$, $\Delta \in (36, 40) > 0$, yielding roots $x = 1 \pm \sqrt{k + 5}$.

Since $9 < k + 5 < 10$, $3 < \sqrt{k + 5} < \sqrt{10} \approx 3.162$:

- The larger root satisfies $x = 1 + \sqrt{k + 5} > 4 > 2$ (valid on $x > 2$).
- The smaller root satisfies $x = 1 - \sqrt{k + 5} < -2$ (valid on $x < -2$).

Thus, there are always exactly 2 distinct solutions on the outer branches.

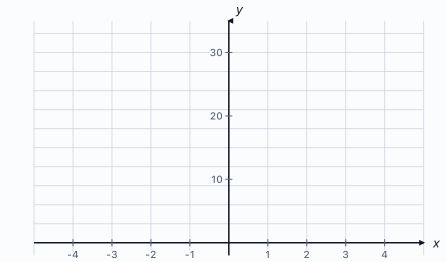
Combining both parts, the equation has $2 + 2 = 4$ distinct real solutions if and only if $4 < k < 5$.

Common Misconceptions:

- $4 \leq k \leq 5$ incorrectly includes the boundary values $k = 4$ (corner intersection) and $k = 5$ (tangency), each of which yields only 3 distinct solutions.
- $-4 < k < 4$ yields only 2 distinct solutions because one root from the inner arch and one root from the outer branch fall outside their valid domains. [M1][A1]

11 MCQ

In the plane, a parabola is given by $f(x) = 3(x-7)^2 + 17$. Find the vertical intercept $f(0)$.



A 171

B 147

C 164 ✓

D 167

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Setting $x = 0$ gives $f(0) = 3(-7)^2 + 17 = 164$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 164. **[A1]**

12 MCQ

Find the set of values of the constant m for which the equation

$$|2x-6| = mx+2$$

has exactly two distinct real solutions.

A $m < -\frac{2}{3}$ or $m > 2$

B $-\frac{2}{3} \leq m \leq 2$

C $-2 < m < 2$

D $-\frac{2}{3} < m < 2$ ✓

WORKED SOLUTION & VERIFICATION

We consider the function $y = |2x-6|$ and the line $y = mx+2$.

1. **Analyze the modulus function:**

The graph of $y = |2x-6|$ consists of two linear rays meeting at the vertex $(3,0)$:

$$y = \begin{cases} 2x-6, & \text{for } x \geq 3 \text{ (gradient } 2), \\ -2x+6, & \text{for } x < 3 \text{ (gradient } -2). \end{cases}$$

2. **Analyze the line $y = mx+2$:**

The line has a fixed y-intercept at $(0,2)$ and variable gradient m.

3. **Lower boundary (passing through the vertex):**

The line passes through the vertex $(3,0)$ when:

$$0 = m(3)+2 \implies m = -\frac{2}{3}.$$

- If $m < -\frac{2}{3}$, the line passes below the vertex $(3,0)$ and does not intersect the right ray $x \geq 3$.
- At $m = -\frac{2}{3}$, the line meets the vertex $(3,0)$ as its only intersection point on both rays, giving exactly one real solution.
- Therefore, for two distinct solutions, we must have $m > -\frac{2}{3}$.

4. **Upper boundary (parallelism with the right ray):**

The right ray has gradient 2.

- If $m = 2$, the line $y = 2x+2$ is parallel to $y = 2x-6$ and never intersects the right ray; it intersects only the left ray at $-2x+6 = 2x+2 \implies x = 1$ (1 solution).
- If $m > 2$, the line is steeper than the right ray and starts above it at $x = 0$, so it only intersects the left ray once and never intersects the right ray.
- Therefore, the line must be less steep than the right ray: $m < 2$.

5. **Verification of two intersections for $-\frac{2}{3} < m < 2$:**

- Intersection with right ray: $2x-6 = mx+2 \implies (2-m)x = 8 \implies x = \frac{8}{2-m}$. Since $m < 2$ and $m > -\frac{2}{3}$, we have $2-m < \frac{8}{3}$, which implies $x > 3$ (valid).
- Intersection with left ray: $-2x+6 = mx+2 \implies (m+2)x = 4 \implies x = \frac{4}{m+2}$. Since $m > -\frac{2}{3} > -2$, $m+2 > \frac{4}{3}$, which implies $0 < x < 3$ (valid).

Thus, exactly two distinct real solutions occur when $-\frac{2}{3} < m < 2$.

Common Traps:

- Setting $m > -2$ (Choice B) mistakenly uses the slope of the left branch instead of testing when the line crosses the vertex $(3,0)$.
- Squaring both sides blindly introduces extraneous roots where $mx+2 < 0$.
- Including the endpoints (Choice D) fails to recognize that at $m = -\frac{2}{3}$ and $m = 2$ there is only one real solution.

[M1][A1]

13 MCQ

The function f is defined for $x \in \mathrm{R}$ by

$$f(x) = x^2 - 2kx + (3k + 10),$$

where k is a real constant, and the equation $f(x) = 0$ has no real roots.
The function g is defined for $x \in \mathrm{R}$ by

$$g(x) = |x - k|.$$

Find the set of values of k for which the minimum value of the composite function $h(x) = f(g(x))$ is strictly greater than 4.

A $\frac{3-\sqrt{33}}{2} < k < \frac{3+\sqrt{33}}{2}$

B $-2 < k < \frac{3+\sqrt{33}}{2}$ ✓

C $-2 < k < 5$

D $0 \le k < \frac{3+\sqrt{33}}{2}$

WORKED SOLUTION & VERIFICATION

We solve the problem in three systematic stages:

1. **Condition from the Discriminant of $f(x) = 0$:**

Since $f(x) = 0$ has no real roots, its discriminant Δ must be strictly negative:

$$\Delta = (-2k)^2 - 4(1)(3k + 10) = 4k^2 - 12k - 40 < 0.$$

Dividing by 4:

$$k^2 - 3k - 10 < 0 \iff (k - 5)(k + 2) < 0 \iff -2 < k < 5.$$

2. **Minimum of the Composite Function $h(x) = f(g(x))$:**

Let $u = g(x) = |x - k|$. Because $x \in \mathrm{R}$, the range of g is $[0, \infty)$.

We express $f(u)$ in completed square form:

$$f(u) = u^2 - 2ku + (3k + 10) = (u - k)^2 + 3k + 10 - k^2.$$

The vertex of this upward-opening parabola is at $u = k$. The location of the vertex relative to the restricted domain $u \ge 0$ depends on the sign of k:

- **Case 1:** $k \ge 0$

The vertex $u = k$ lies within the domain $[0, \infty)$. Hence, the minimum value is attained at $u = k$:

$$f(k) = -k^2 + 3k + 10.$$

We require $f(k) > 4$:

$$-k^2 + 3k + 10 > 4 \iff k^2 - 3k - 6 < 0.$$

The roots of $k^2 - 3k - 6 = 0$ are $k = \frac{3 \pm \sqrt{33}}{2}$. Since $k \ge 0$ and $\frac{3+\sqrt{33}}{2} \approx 4.372 < 5$, this gives:

$$0 \le k < \frac{3 + \sqrt{33}}{2}.$$

- **Case 2:** $k < 0$

The vertex $u = k$ lies strictly to the left of $[0, \infty)$. Since the parabola is strictly increasing for $u \ge k$, its minimum on $[0, \infty)$ occurs at the boundary $u = 0$:

$$f(0) = 3k + 10.$$

We require $f(0) > 4$:

$$3k + 10 > 4 \iff 3k > -6 \iff k > -2.$$

Combining with $k < 0$, this yields $-2 < k < 0$.

3. **Combining the Conditions:**

Combining Case 1 and Case 2 gives:

$$(-2, 0) \cup [0, \frac{3 + \sqrt{33}}{2}) = -2 < k < \frac{3 + \sqrt{33}}{2}.$$

Since $\frac{3+\sqrt{33}}{2} \approx 4.372 < 5$, this entire interval lies within the initial constraint $-2 < k < 5$.

Common Misconceptions:

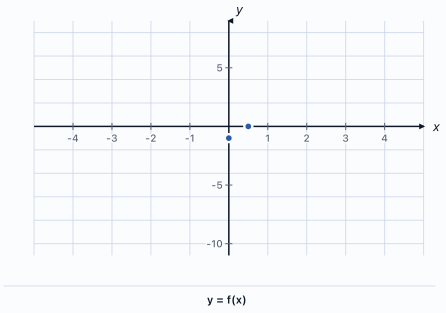
- Choice B incorrectly assumes that the vertex $u = k$ is always reachable for all k, failing to recognize that $u = |x - k| \geq 0$ prevents u from reaching k when $k < 0$.
- Choice C mistakenly assumes the minimum always occurs at $u = 0$.
- Choice D overlooks the valid negative values of $k \in (-2, 0)$. [M1][A1]

14 MCQ

The functions f and g are defined for all real values of x by

$$f(x) = 2x - 1 \quad \text{and} \quad g(x) = \frac{x+5}{3}.$$

Find the value of x such that $fg(x) = 9$.



A $x = 9$

B $x = 5$

C $x = \frac{23}{2}$

D $x = 10$ ✓

WORKED SOLUTION & VERIFICATION

To find x such that $fg(x) = 9$, we can either determine the composite expression $fg(x)$ or solve sequentially from the outer function to the inner function.

Method 1 (Sequential Evaluation):

We have $f(g(x)) = 9$.

Since $f(u) = 2u - 1$, setting $f(u) = 9$ gives:

$$2u - 1 = 9 \implies 2u = 10 \implies u = 5.$$

Thus, $g(x) = 5$.

Now solve for x:

$$\frac{x+5}{3} = 5 \implies x + 5 = 15 \implies x = 10.$$

Method 2 (Forming the Composite Function):

$$fg(x) = f(g(x)) = 2\left(\frac{x+5}{3}\right) - 1 = \frac{2x+10}{3} - 1 = \frac{2x+7}{3}.$$

Setting this equal to 9:

$$\frac{2x+7}{3} = 9 \implies 2x + 7 = 27 \implies 2x = 20 \implies x = 10.$$

Analysis of Distractors:

- Choice B ($\frac{23}{2}$) is obtained by evaluating the reverse composite function $gf(x) = 9$, where $\frac{(2x-1)+5}{3} = 9 \implies 2x + 4 = 27 \implies x = \frac{23}{2}$.
- Choice C (9) arises from an algebraic error when clearing the denominator without multiplying the constant term -1 by 3: $2(x+5) - 1 = 27 \implies 2x + 9 = 27 \implies x = 9$.
- Choice D (5) results from stopping early after finding the intermediate value $g(x) = 5$ (or mistakenly solving $f(x) = 9$). [M1][A1]

15 MCQ

The quadratic equation $x^2-(2k+1)x+(k^2+2)=0$ has real roots α and β. Given that $\alpha^2+\beta^2=27$, find the value of the constant k.

- **A** 3 ✓
- **B** -5
- **C** 3 or -5
- **D** 5

WORKED SOLUTION & VERIFICATION

From the relations between roots and coefficients for the quadratic equation $x^2-(2k+1)x+(k^2+2)=0$:

$$\alpha+\beta=2k+1 \quad \text{and} \quad \alpha\beta=k^2+2$$

Expressing $\alpha^2+\beta^2$ in terms of the sum and product of the roots:

$$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$$

Substitute the known values:

$$(2k+1)^2-2(k^2+2)=27$$

Expand and simplify:

$$(4k^2+4k+1)-(2k^2+4)=27$$

$$2k^2+4k-3=27 \implies 2k^2+4k-30=0 \implies k^2+2k-15=0$$

Factoring the quadratic equation:

$$(k-3)(k+5)=0 \implies k=3 \quad \text{or} \quad k=-5$$

A common algebraic trap is to accept both solutions without checking the condition for real roots. For the equation to have real roots, the discriminant Δ must be non-negative ($\Delta \geq 0$):

$$\Delta=[-(2k+1)]^2-4(1)(k^2+2)=4k^2+4k+1-4k^2-8=4k-7\geq 0 \implies k\geq\frac{7}{4}$$

Checking each value:
- For $k=3$: $4(3)-7=5>0$, so the roots are real.
- For $k=-5$: $4(-5)-7=-27<0$, which produces non-real complex roots.

Thus, the only valid value of the constant is $k=3$. [M1][A1]

16 MCQ

In this problem, the function $f(x)=\frac{4x+1}{x-2}$ is defined for $x\neq 2$. Find the inverse function $f^{-1}(x)$.

- **A** $f^{-1}(x)=\frac{2x-1}{x-4}$
- **B** $f^{-1}(x)=\frac{2x+1}{x-4} \quad (x\neq 4)$ ✓
- **C** $f^{-1}(x)=\frac{x-2}{4x+1}$
- **D** $f^{-1}(x)=\frac{4x-2}{x+1}$

WORKED SOLUTION & VERIFICATION

Let $y=\frac{4x+1}{x-2} \implies y(x-2)=4x+1 \implies yx-2y=4x+1 \implies x(y-4)=2y+1 \implies x=\frac{2y+1}{y-4}$. Swap variables to obtain $f^{-1}(x)=\frac{2x+1}{x-4}$. [M1][A1]

17 MCQ

Observe that the function $f(x) = \frac{5x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{x-2}{5x+1}$

B $f^{-1}(x) = \frac{2x+1}{x-5} \quad (x \neq 5)$ ✓

C $f^{-1}(x) = \frac{5x-2}{x+1}$

D $f^{-1}(x) = \frac{2x-1}{x-5}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{5x+1}{x-2} \implies y(x-2) = 5x+1 \implies yx-2y = 5x+1 \implies x(y-5) = 2y+1 \implies x = \frac{2y+1}{y-5}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-5}$. [M1][A1]

18 MCQ

Suppose the function $f(x) = \frac{6x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{x-2}{6x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-6}$

C $f^{-1}(x) = \frac{2x+1}{x-6} \quad (x \neq 6)$ ✓

D $f^{-1}(x) = \frac{6x-2}{x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{6x+1}{x-2} \implies y(x-2) = 6x+1 \implies yx-2y = 6x+1 \implies x(y-6) = 2y+1 \implies x = \frac{2y+1}{y-6}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-6}$. [M1][A1]

19 MCQ

Notice that the function $f(x) = \frac{7x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{7x-2}{x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-7}$

C $f^{-1}(x) = \frac{x-2}{7x+1}$

D $f^{-1}(x) = \frac{2x+1}{x-7} \quad (x \neq 7)$ ✓

WORKED SOLUTION & VERIFICATION

Let $y = \frac{7x+1}{x-2} \implies y(x-2) = 7x+1 \implies yx-2y = 7x+1 \implies x(y-7) = 2y+1 \implies x = \frac{2y+1}{y-7}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-7}$. [M1][A1]

20 MCQ

Consider the function $f(x) = \frac{8x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x+1}{x-8} \quad (x \neq 8)$ ✓

B $f^{-1}(x) = \frac{x-2}{8x+1}$

C $f^{-1}(x) = \frac{2x-1}{x-8}$

D $f^{-1}(x) = \frac{8x-2}{x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{8x+1}{x-2} \implies y(x-2) = 8x+1 \implies yx-2y = 8x+1 \implies x(y-8) = 2y+1 \implies x = \frac{2y+1}{y-8}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-8}$. [M1][A1]

21 MCQ

In this problem, the function $f(x) = \frac{9x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{9x-2}{x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-9}$

C $f^{-1}(x) = \frac{2x+1}{x-9} \quad (x \neq 9)$ ✓

D $f^{-1}(x) = \frac{x-2}{9x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{9x+1}{x-2} \implies y(x-2) = 9x+1 \implies yx-2y = 9x+1 \implies x(y-9) = 2y+1 \implies x = \frac{2y+1}{y-9}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-9}$. [M1][A1]

22 MCQ

Solve the modulus equation $|2x+6| = 4$.

A $x = -5$ or $x = -1$ ✓

B $x = -3$ or $x = 1$

C $x = 5$ or $x = 1$

D $x = -1$ only

WORKED SOLUTION & VERIFICATION

Split into two linear cases: $2x+6 = 4 \implies x = -1$, and $2x+6 = -4 \implies x = -5$.

23 MCQ

Solve the modulus equation $|3x+6| = 6$.

A $x = -2$ or $x = 2$

B $x = -4$ or $x = 0$ ✓

C $x = 4$ or $x = 0$

D $x = 0$ only

WORKED SOLUTION & VERIFICATION

Split into two linear cases: $3x+6 = 6 \implies x = 0$, and $3x+6 = -6 \implies x = -4$.

24 MCQ

Solve the modulus equation $|4x+4| = 8$.

A $x = -3$ or $x = 1$ ✓

B $x = 3$ or $x = -1$

C $x = -1$ or $x = 3$

D $x = 1$ only

WORKED SOLUTION & VERIFICATION

Split into two linear cases: $4x+4 = 8 \implies x = 1$, and $4x+4 = -8 \implies x = -3$.

25 MCQ

Solve the modulus equation $|5x| = 10$.

A $x = 2$ or $x = -2$

B $x = 2$ only

C $x = 0$ or $x = 4$

D $x = -2$ or $x = 2$ ✓

WORKED SOLUTION & VERIFICATION

Split into two linear cases: $5x = 10 \implies x = 2$, and $5x = -10 \implies x = -2$.